

STEAMROOT · REQUIREMENTS

SteamRoot requirements

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DOCUMENT CONTROL

Document ID	STR-REQ-001	Version	0.3
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/steamroot		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	Apply Amish's 2026-09-25 decisions (two hoods in R2, \$1,800 budget in R12); correct the wood heating value in R4; add TRL 3 status from STR-CAL-001
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2, including pressure and safety requirements
0.1	2026-09-24	Amish Chadha	Initial scaffold

SteamRoot requirements

These requirements were checked by calculation at TRL 3 in STR-CAL-001 (docs/04-calcs/01-sizing.md). Seven of the fourteen checks are not met on paper; the targets are unchanged except where Amish decided otherwise on 2026-09-25 (STR-DDR-001). Safety requirements (R6, R7, R9 and R10) take priority over performance requirements if they conflict.

Safety: SteamRoot is a fired steam generator. Meeting these requirements on paper does not make it safe or legal to build. A written ruling from the local boiler authority is an open prerequisite for any detail design or build.

Table 1. Requirements and TRL 3 status.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (STR-CAL-001)
R1	Pasteurize soil to working depth	Soil at 70 °C (158 °F) or more for 20 to 30 min at 10 to 15 cm depth across at least 80 % of the hood footprint	Heat balance calculation; later thermocouple probes at 15 cm	Not verifiable at TRL 3 (1D model holds 70 °C at 15 cm if the front reaches 16.5 cm)
R2	Treat beds at a useful rate	2 m ² /h or more at 15 cm depth; 5 m ² /h or more at 5 cm depth (weed seed bank only), with two hoods used alternately (decided by Amish, 2026-09-25)	Calculation from steam output, steam per m ² and hood cycle	Not met: 1.88 m ² /h at 15 cm, 3.41 m ² /h at 5 cm
R3	Generate enough steam	30 kg/h or more of saturated steam, continuous, from one firebox	Heat transfer and combustion calculation	Met: 30 kg/h at 40.7 kW firing
R4	Limit fuel use	4 kg or less of wood at 20 % moisture (wet basis; about 14.5 MJ/kg lower heating value) per m ² at 15 cm depth	Energy balance calculation	Not met: 5.4 kg/m ²
R5	Use biomass efficiently	65 % or more of fuel energy delivered as steam, with the economizer	Energy balance calculation; later flue temperature and fuel weighing	Not met: 54 %
R6	Stay near atmospheric pressure	Normal working pressure 0.1 bar gauge (1.5 psi) or less at every point on the steam side; an open vent to atmosphere that cannot be closed or isolated	Pressure drop calculation; design review	Not met: 0.110 bar at the coil inlet (header 0.033 bar); open vent met by design
R7	Keep stored energy low	8 L or less of water in the heated section; no closed steam drum	Volume calculation from the coil geometry	Met: 7.0 L flooded
R8	Run between refills	3 h or more of steaming per feed water fill	Tank volume and steam rate	Met: 4.2 h
R9	Protect against overpressure	A certified steam relief valve, sized for full steam output and set at the lowest available certified pressure, 1 bar (15 psi) or less, as a second line of protection behind R6	Valve rating and capacity check	At risk: the lowest standard set, 15 psi, is 1.03 bar; wording fix proposed
R10	Protect the operator and site	Chimney outlet 2.2 m or more above ground with a spark arrestor (mesh 6 mm or finer); fire outdoors only; steam diverted to the vent, not the hood, before the hood is lifted; hood handles 60 °C or cooler	Design review; later surface temperature check	Met: 2.4 m, 6 mm mesh, hood skin 23 °C
R11	Tow with farm vehicles	Loaded mass 500 kg or less with a full tank; overall width 1.5 m or less	Mass estimate from the model and BOM	Not met: 536 kg. Width at risk: 1.48 m
R12	Build within the prototype budget	Parts cost \$1,800 or less (raised from \$1,500 by Amish on 2026-09-25 to allow about 20 % contingency)	Priced BOM	Not met: \$1,915

■ Assumptions

- Soil to 15 cm depth: bulk density 1,300 kg/m³, gravimetric water content 20 % (dry basis), starting temperature 15 °C. At TRL 3 the soil is modeled with a condensation front: soil above the front reaches 100 °C, and the front must reach 16.5 cm so that 15 cm stays at 70 °C or more through a 25-minute hold (STR-CAL-001, section 2). The TRL 2 model heated a uniform layer to 75 °C.
- Specific heat of dry mineral soil 0.84 kJ/(kg K), of water 4.18 kJ/(kg K).
- 70 °C for 30 minutes is widely used as a soil pasteurization target for most weed seeds and fungal pathogens. Some hard-seeded weeds and heat-tolerant viruses need higher temperatures; this is accepted.
- Wood at 20 % moisture (wet basis) has a lower heating value of about 14.5 MJ/kg for a typical hardwood analysis. Earlier versions quoted about 16 MJ/kg, which was too high.
- Pressure thresholds for boiler and pressure vessel rules vary by jurisdiction. R6 and R9 are set to be well below common low-pressure thresholds, but compliance must be confirmed locally. The written ruling from the local boiler authority is an open prerequisite.
- The R2 treatment rate assumes two hoods used alternately, a 25-minute hold, 2 minutes to reset a hood and 1 minute of vented steam each time the hose is moved.